

Flood Hazards and Urban Housing Markets: The Effects of Katrina on New Orleans

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Abstract This study examines the impacts on consumers' willingness to pay for certain characteristics of housing in greater New Orleans before and after the flooding of Hurricane Katrina. Single-family home sales from January 2004 to August 2006 are collected and used in a hedonic price function to estimate the changes in the value of amenities, and structural, neighborhood and geographic characteristics, including the mean elevation of each property. Elevation, which buyers did not know for certain prior to the storm, but may now be inferred from water level marks in most neighborhoods, is found to have a positive relationship with selling prices. Results indicate that pre-Katrina, there was a premium of only 1.4% per foot in flood-prone areas, and was insignificant in areas not subject to flooding. This increased to 4.6% for flooded areas after Katrina. These findings are attributed to not only the perceived *risk* of flooding, but also to the potential of higher compliance costs associated with rebuilding under more stringent National Flood Insurance Program (NFIP) guidelines.

Keywords Housing markets · Hedonic estimation · Flooding · Elevation · Housing supply · Katrina

Introduction

The landfall of Hurricane Katrina on August 29, 2005 and the flooding that followed devastated the city of New Orleans. With the breaching of the levees protecting New

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Orleans from Lake Ponchartrain, approximately 80% of the city was flooded. While the city's population remains at roughly 60% of its pre-storm level (Brookings 2007), its housing stock may be less than that. Estimates suggest that, in some areas, more than one-half of the housing units were either severely damaged or destroyed. According to a U.S. Housing and Urban Development (HUD) study (2006), approximately 34,300 (19.5% of the pre-storm stock) of the units in Jefferson Parish suffered major or severe damage. In Orleans Parish, the number experiencing this level of damage was 105,300, or 56% of the pre-storm units. Since the storm, many of these have been demolished, while others stand in ruin.

With slightly more than half of its original population and as much as 38% of its original housing stock destroyed, significant market adjustments are occurring to ensure market clearing outcomes. According to a Brookings Institute study (2006), the number of Orleans Parish houses on the market has risen from roughly 2,800 in February 2006 to nearly 5,000 in October. During this same period, housing sales increased only from 190 to 291. After a sharp spike immediately following the storm, average housing prices retreated to approximately \$222,000 in February 2006, significantly more than the pre-storm average of \$203,000. Between February 2006 and February 2007, average prices fell to slightly more than \$197,000, well below the pre-storm level (MLS). These price changes may well reflect changes in housing quality after the storm. If a substantial portion of the stock of housing was damaged and not repaired, lower average prices would be expected.

From an economic perspective, these events provide an opportunity to examine the responses of individuals in the local housing market to the realization of a major flood. This paper estimates a hedonic price equation for a large sample of homes in New Orleans to determine the effect of Katrina on the willingness-to-pay for various amenities and housing characteristics. The uniqueness of this study is the treatment of flood hazard and the incorporation of Geographic Information Systems (GIS) data to extract information about individual property and neighborhood characteristics and flooding in New Orleans. Previous studies using floodzone classifications have yielded step-wise relationships between home prices and distinct classifications. The current research produces a continuous gradient of home prices relative to elevation. Two sets of regressions are conducted to determine these values both before and after Katrina. This allows an examination of how consumer's willingness-to-pay has changed as a result of the storm.

Review of Hedonic Estimation and Flood Risk

It is commonly accepted that housing is a composite good, in that it is made up of many distinct characteristics. Through a process known as hedonic estimation [Wallace 1926, Court 1939, and Rosen 1974, among others], economists are able to estimate the contribution to the total value of real estate attributable to each characteristic possessed by that property. In general, the characteristics that have been shown to have significant impacts on the price of a house include size, number of bedrooms, number of bathrooms, and car storage, among others. While the structural features of the house are certainly important, other factors such as neighborhood characteristics have also been shown to have an impact on the price of property.

Many recent studies have examined the impact of flood risk exposure on housing values. Most of these attempt to describe the discount associated with location within a defined floodplain. Bin and Polasky (2004) examine the effects of location in a floodplain in their analysis of housing prices in North Carolina. In addition to numerous structural and locational characteristics, they include a binary variable for flood risk, as determined by whether the property is located in a 100-year flood zone. Their findings suggest that location in a floodplain is significant, and reduces housing prices by 5.8%, or \$7,529 at the mean. They also show that the occurrence of Hurricane Floyd impacted the influence of location in a floodplain by conducting a separate analysis of the data before and after the storm. This shows that prior to Floyd, the discount for location in a floodplain was 3.8%, or \$4,933 at the mean, while after the storm it rose to 8.3%, or \$10,774.

In their study of the effects of disclosure of floodplain status on the value of residential real estate in California, Troy and Romm (2004) find that there is a 4.2% discount associated with a property's location in a floodplain. MacDonald et al. (1990) also find a negative and significant influence of location in a floodplain on housing prices. Their estimates varied with the predicted sales price, with an average priced home (\$65,000 in 1990) subject to a discount of 3.7%.

Speyrer and Ragas (1991) examine the effects of floodplain classification on property values in New Orleans, and find results consistent with those mentioned and significant. Their estimates indicate a maximum discount for location in a floodplain of 8.3%. Finally, Shilling et al. (1985) find that houses located in a floodplain sell at a 6.4% discount relative to comparable houses outside floodplain zones.

Several of these studies were conducted for time periods following major storm events, while others were not. Those conducted within a relatively short time of a flood (Speyrer and Ragas 1991 and Bin and Polasky 2004) find a higher discount for location in a floodplain. Those with no recent flooding show lower discounts. The difference in findings of these studies suggests that consumer willingness-to-pay for a reduction in flood risks increases with recent flooding.

Some studies have found location in a floodplain to be insignificant in determining the price of real estate, such as Zimmerman (1979), Babcock and Mitchell (1980), Skantz and Strickland (1987) and Bialaszewski and Newsome (1990). While the structural and locational variables used in these studies are often different, the consistent aspect is that each uses a binary variable of location in a 100-year floodplain to proxy for flood risk. One possible explanation for the insignificance of the flood risk variable noted in some of the studies is the implicit assumption that the flood risk is constant across the entire floodplain, and thus the discount associated is also constant. In this study, the authors do not make such an assumption.

The study at hand recognizes the likelihood that flood risks and the associated discounts are inversely related to the elevation of the property *within* the floodplain. Some studies (see Bartosova et al. 1999) have attempted to capture a similar measure of risk by including the distance from the potential flood source as a variable in the hedonic price equation. While this approach may well capture more of the risk of flooding, it does not reflect the risk-extent of flooding interaction possible with the use of elevation measures, especially in a flood event such as that which struck New

Orleans. In this case, the flooding occurred as a result of equalization of water levels between a large body of water surrounding a bowl shaped city. Distance from the lake had no bearing on whether or not a certain house flooded. More importantly, the flooding from Katrina was unique in that most floods are either flash-floods, or caused by inadequate drainage; they are moving-water floods. The flooding that followed Katrina was a standing-water flood, so that elevation was of primary importance.

In earlier studies, the justification for the price differentials was the capitalized difference in flood insurance premiums. Therefore, the step-wise function is likely a good representation of prices. However, it assumes that because all homes in a given flood zone pay the same premia, the associated discounts are equal. The approach used in this study avoids this step-wise pricing effect by incorporating a continuous measure of elevation, which for all but flashflood events, better reflects the risks associated from flooding. The result of this approach is a flood-risk discount gradient based on elevation, and irrespective of flood zone classification. In addition, the recent flooding in New Orleans may provide home buyers with a visual cue to the possibility and extent of flood damages. This is due to the water level marks left on many of the homes and buildings. Even now, two years after the storm, water marks on New Orleans homes are ubiquitous.

Data and Study Area

The data used in this study include information regarding terms-of-sale, locational and structural characteristics, and amenities for more than 33,700 single-family homes sold in southeastern Louisiana between January 2004 and August 2006. The data were collected from the Multiple Listing Service (MLS), a database used by local realtors for listing properties. As the application of hedonic estimation implies a homogenous market, where there are no systematic differences in the goods exchanged, only those properties located in Jefferson and Orleans Parishes are considered. Since greater New Orleans is primarily made up of these two parishes, this delineation is used to reduce the degree of heterogeneity in the defined market. The resulting number of observations is 16,258. The time horizon of the data includes 20 months prior to Hurricane Katrina and 18 months after. The authors believe this to be sufficient time for market adjustments to reflect the views of market participants.

The choice of variables to include in the price equation was driven by a review of the hedonic literature. The characteristics that have been shown to significantly influence the price of residential properties are many. While there are differences among studies, some characteristics are consistently shown to be significant. A relatively complete review of the literature is presented by Sirmans et al. (2005), as well as Malzeppi (2003). The variables used in this study are presented in Table 1, with their expected signs and source. For clarity, these are divided into two classifications. While several classification schemes have been offered for grouping these characteristics, Wilkinson (1973) offered possibly the most basic, and useful distinction between the types of factors involved—dwelling-specific vs. location-specific characteristics. This is the distinction used in this study.

Table 1 Variables

Variable	Description	Source	Expected Sign
Dwelling-specific			
Living	Approximate living area (100 ft ²)	MLS	+
Bdrms	Number of bedrooms	MLS	+/-
Hbath	Number of half baths	MLS	+
Fbath	Number of full baths	MLS	+
Age	Age of house in decades	MLS	+/-
A/C	Air conditioning, 1 = central, 0 = otherwise	MLS	+
Fire	1 = has fireplace, 0 = otherwise	MLS	+
Cars	Number of car spaces	MLS	+
Cond	Listing agent's evaluation (scale = 1–5)	MLS	+
Pool	1 = has a pool, 0 = otherwise	MLS	+
Roof	Roof, 1 = flat, 0 = otherwise	MLS	–
ToMkt	Time on market (months)	MLS	–
Location-specific			
Wf	Location, 1 = waterfront, 0 = otherwise	MLS	+
Lot	Lot size in 1,000's of square feet	MLS	+
Race	Census tract % white	Census Bureau	+
MHI	Census tract median household income (\$1,000's)	Census Bureau	+
CBD	Distance to centroid of CBD (in miles)	ArcGis	–
Elev	Elevation of centroid of property (feet above MSL)	ArcGis	+

The dwelling-specific characteristics used in this study reflect the type, size, condition, and amenities associated with each property. They include the number of bedrooms, full bathrooms, half bathrooms, and total living area. Other dwelling-specific variables include an assessed condition of the property, car parking arrangements, type of roof, age of property, lot size, the presence of a pool, and type of air conditioning. Finally, for convenience, a market condition variable, days-on-market, is also included in dwelling-specific characteristics.

The included assessed condition variable is especially important to this exercise, as the condition of much of the housing in post-Katrina New Orleans has changed relative to that of pre-Katrina. The state of repair (disrepair) of the housing stock is reflected in this variable. The source of this data is MLS. Listing agents assign a condition classification to each listed property, based on an established set of criteria. The meanings and implications of the assessed condition are given below:

0. *poor*—excessive abuse evident, abandonment or major reconstruction needed
1. *fair*—badly worn, in need of major overhaul, or post-Katrina gutted
2. *average*—major components still functional, minor repairs needed
3. *very good*—no obvious maintenance required, appearance and utility above average
4. *excellent*—no functional inadequacies, like-new condition
5. *new*—new construction, never occupied

By controlling for condition, we ensure that the lower price of the houses at lower elevations is due to the fact that they are damaged. That is, we control for condition so that the coefficient on elevation better captures future flood risk, rather than present flood damage.

As a result of widespread flooding, and the damages incurred, many properties have been partially or completely gutted prior to sale. Gutted houses are stripped of all interior features, including drywall, fixtures, and insulation, often leaving only the skeleton (the two-by-fours) in tact. This gutting significantly changes the characteristics of the property, and is reflected in the assessed condition variable by a lower score, as compared to an otherwise usable house. A gutted home that was not flooded takes on a lower assessed value; however, gutting can be an improvement over a moldy flooded home. This is due to the fact that gutting is the first step in rehabilitating these structures, so some of the required work has already been done. Gutting allows the skeleton of the house to dry, and helps prevent the growth of mold.

With the exception of waterfront location, the locational-specific characteristics used in this study were extracted from multiple sources using Geographic Information Systems (GIS) tools. Each property was geo-referenced, or coded for mapping, using ArcView GIS software. This process uses a digital map of the area including streets and roads to locate a property based on its address. While some properties can be coded, others cannot, due to incorrect or misspelled addresses. As each property is located, it is assigned a score reflecting the likelihood the location is accurate, with 100 being a perfect score. Of the properties in the data, 2,144 could not be coded. From those that were successfully coded, only those with scores above 75 were used. While the default minimum match score in ArcInfo is 60%, setting a higher threshold increases the likelihood of accurate matches. The resulting sample size was 11,617 properties, including omissions for incomplete data. Nine homes (one in Orleans and eight in Jefferson Parish) closed in the month immediately following the storm, on terms that would have been agreed to prior to the storm. These nine observations are thus assigned to the pre-Katrina subset.

Once the observations were coded, different map layers were used to extract additional data for each location. Parish-level mosaics of the LiDAR data from U.S. Geological Survey (USGS) were used to extract the elevation of each property, relative to mean sea level (msl). U.S. Census Bureau Tiger files were used for census tract-level demographics, including median household income (MHI) and racial composition. Finally, ArcView geospatial tools were used to calculate the Euclidean distance to the centroid of the central business district (CBD).

Clearly the housing market in greater New Orleans (GNO; which we define here to include Jefferson and Orleans parishes) is different pre-and post-Katrina. Thus, we estimate each sample separately. Special care should be made when describing the post-Katrina real estate market in GNO. A question of heterogeneity arises between flooded and non-flooded houses. Should flooded and non-flooded homes be lumped together into a single post-Katrina regression, or should they be separated?

To separate the samples, those houses within the post-Katrina data that were flooded due to Katrina and those that were not must be identified. This was accomplished by delineating the areas that were flooded using flood-depth maps from the Federal Emergency Management Agency (FEMA). Since the flooding was the result of breaches in levee protection and the resultant equalization of surface water levels, elevation was the determining factor. Only those areas above sea level and those for which levees did not fail were spared.

All houses located in areas where flooding was indicated by FEMA maps and built on concrete slab foundations were assigned to the flooded subset. For homes on

raised foundations, a question of flooding arises in those areas receiving only small amounts of water. Visual inspection of a sample of homes reveals that most raised foundations vary between two and four feet. As such, it is not possible, with any certainty, to systematically assign these raised houses to one subset or the other. Thus, only those homes built on concrete slabs are included in this study.

There are systematic differences between houses in areas that flooded relative to those that did not, not only regarding elevation, but also in the housing and neighborhood's socio-economic characteristics. These differences suggest that applying the same bifurcation to the pre-Katrina sample that was used in the post-Katrina sample would ensure that comparisons between pre- and post-Katrina regressions were consistent. Applying this bifurcation process to both samples ensured that comparisons between pre- and post-Katrina regressions are consistent. The result of this process is a dataset consisting of four distinct subsets: pre-Katrina flooded and non-flooded, and post-Katrina flooded and non-flooded. Summary statistics are presented in Table 2.

Houses in low-lying areas tended to be in poor condition, and inhabited by low income minority residents. Pre-Katrina, the assessed condition in the flood areas has a mean of 3.26, while the non-flood area had a mean of 3.44. Post-Katrina flooded homes had a slightly lower mean condition of 2.81, while non-flooded homes show no change in their mean of 3.44. While the non-flood sample mean doesn't change post-Katrina, there is a significant decline in the flooded sample's mean. It seems reasonable to attribute this decline to the effects of the storm.

Methodology

Economic theory does not specify which variables should be included in a hedonic regression, nor does it specify the functional form, unless the precise aggregate utility function is known in advance. The matter is partially one of fit, choosing the functional form that best describes the data. The choice of functional forms has been studied by Halvorsen and Pollakowski (1981), Early and Sinclair (1983), and Cropper et al. (1988), with no clear consensus regarding a preferred approach. Researchers commonly experiment with various functional forms, most commonly the Box–Cox transformations: linear, log–linear, log–log. This study applies the following log–linear form:

$$\ln \text{Price} = f(\text{elev}, \text{cbd}, \text{living}, \text{bdrms}, \text{fbath}, \text{hbath}, \text{age}, \text{a/c}, \text{fire}, \text{cars}, \text{cond}, \text{pool}, \text{roof}, \text{tomkt}, \text{wf}, \text{lot}, \text{race}, \text{lnmhi})$$

where the variable descriptions and expected signs are as given in Table 1. The primary reason for the choice of a semi-log price equation is to allow for non-linear relationships that have proven to be superior in many other empirical studies. Log–linear models also minimize the influence of heteroskedasticity.

The housing market in New Orleans following the storm was in a state of flux, with a higher than usual variance in home prices. To capture the increased turbulence of the local market, we include a series of dummy variables, one for every month in our sample. This is akin to accounting for the effects of time semi-parametrically. Such terms simultaneously capture seasonality as well as any possibly non-linear trends

Table 2 Summary statistics

Variable	Obs	Mean	Std.Dev.	Min	Max
Pre-Katrina, non-flooded					
Price	3,050	\$201,191	151,442	12,500	980,000
Elev	2,005	2.00	3.74	-8	24
CBD	2,005	6.18	2.43	0	18
Bdrms	3,050	3.32	0.73	0	9
Hbath	3,050	0.37	0.54	0	5
Fbath	3,050	1.99	0.64	0	8
Age(decades)	2,985	2.78	2.19	0	18
Cond	3,050	3.44	1.12	0	5
Lot(1,000 ft ²)	2,769	75.34	148.05	6	5,249
Pool	3,050	0.10	0.30	0	1
Roof	3,050	0.00	0.07	0	1
A/C	3,050	0.91	0.29	0	1
ToMkt	3,050	3.40	2.69	0	36
Living(100 ft ²)	3,042	19.48	8.51	0	96
Cars	3,050	1.12	0.99	0	3
Fire	3,050	0.35	0.48	0	1
Wf	3,050	0.01	0.11	0	1
MHI	2,004	45.58	20.52	6	146
Race	2,005	0.66	0.27	0	1
Pre-Katrina flooded					
Price	4,928	\$176,656	109,952	7,500	970,000
Elev	4,928	-4.68	2.15	-14	0
CBD	4,928	7.46	2.78	1	14
Bdrms	4,928	3.29	0.66	0	8
Hbath	4,928	0.31	0.50	0	5
Fbath	4,928	1.94	0.58	0	6
Age(decades)	4,847	2.97	1.41	0	12
Cond	4,928	3.26	1.12	0	5
Lot(1,000 ft ²)	4,562	65.39	24.81	3	653
Pool	4,928	0.10	0.30	0	1
Roof	4,928	0.00	0.07	0	1
A/C	4,928	0.91	0.29	0	1
ToMkt	4,928	3.02	2.86	0	126
Living(100 ft ²)	4,926	18.59	7.01	0	66
Cars	4,928	1.04	0.98	0	3
Fire	4,928	0.32	0.47	0	1
Wf	4,928	0.00	0.06	0	1
MHI	4,926	44.78	13.20	9	110
Race	4,928	0.59	0.33	0	1
Post-Katrina, non-flooded					
Price	1,602	\$230,317	153,562	10,000	962,500
Elev	1,079	2.00	3.70	-7	13
CBD	1,079	6.29	2.24	1	18
Bdrms	1,602	3.37	0.70	0	7
Hbath	1,602	0.36	0.53	0	5
Fbath	1,601	2.06	0.62	0	5
Age(decades)	1,582	2.70	2.05	0	20
Cond	1,602	3.44	1.05	0	5
Lot(1,000 ft ²)	1,419	73.22	63.49	0	1,693
Pool	1,602	0.11	0.31	0	1
Roof	1,602	0.00	0.04	0	1
A/C	1,602	0.95	0.21	0	1
ToMkt	1,602	3.28	2.39	0	24
Living(100 ft ²)	1,601	19.98	8.15	7	65

Table 2 (continued)

Variable	Obs	Mean	Std.Dev.	Min	Max
Cars	1,602	1.25	0.96	0	3
Fire	1,602	0.39	0.49	0	1
Wf	1,602	0.01	0.09	0	1
MHI	1,079	47.58	20.49	17	146
Race	1,079	0.71	0.23	0	1
Post-Katrina flooded					
Price	2,037	\$196,020	100,071	10,500	905,000
Elev	2,037	-4.33	1.98	-13	0
CBD	2,037	7.50	2.82	2	14
Bdrms	2,037	3.32	0.66	0	7
Hbath	2,037	0.30	0.50	0	3
Fbath	2,037	2.04	0.59	0	5
Age (decades)	2,013	3.07	1.41	0	10
Cond	2,037	2.81	1.38	0	5
Lot (1,000 ft ²)	1,845	67.28	25.09	16	666
Pool	2,037	0.09	0.29	0	1
Roof	2,037	0.00	0.07	0	1
A/C	2,037	0.93	0.26	0	1
ToMkt	2,037	2.89	2.06	0	24
Living (100 ft ²)	2,031	19.66	7.14	0	62
Cars	2,037	1.17	0.97	0	3
Fire	2,037	0.33	0.47	0	1
Wf	2,037	0.00	0.06	0	1
MHI	2,036	49.32	15.43	17	110
Race	2,037	0.75	0.24	0	1

within in the overall market. This is provides an automated way of computing a local house price index, and deflating each house by the average monthly level of prices.

Regarding the treatment of the flooded and non-flooded subsets, whether the values of characteristics are independent of flooding can be revealed by computing separate regressions on the two samples using the method of Seemingly Unrelated Regressions. A series of chi-squared tests then verify whether the coefficients differ across samples. Referring to columns (1) and (2) in Table 3, in the pre-Katrina sample, roughly half of the estimated coefficients are different between the flooded and non-flooded sub-samples. We find the same effect the post-Katrina sample, where roughly half the reported coefficients are statistically different between the flooded and non-flooded sub-samples. This justifies our bifurcation of the dataset along flooded and non-flooded lines.¹

Empirical Results

Estimates for the hedonic price equation coefficients are presented in Table 4. The first two columns report regression results in the market for housing characteristics

¹ We thank an anonymous referee for suggesting this double bifurcation along pre- and post-Katrina, and flooded vs non-flooded lines.

Table 3 Testing for coefficient equality

Variables	<i>P</i> -value			
	Flooded vs. non-flooded		Pre-K vs. post-K	
	Pre-K ^a	Post-K ^b	Flooded ^c	Non-flooded ^d
Elevation	0.03	0.00	0.00	0.91
Distance to CBD	0.98	0.98	0.27	0.25
Bedrooms	0.06	0.48	0.28	0.15
Half-baths	0.01	0.04	0.10	0.54
Full-baths	0.91	0.03	0.16	0.33
Age	0.01	0.67	0.21	0.08
Age squared	0.04	0.82	0.42	0.41
Condition	0.60	0.00	0.03	0.18
Lot area	0.00	0.74	0.15	0.00
Pool	0.95	0.03	0.13	0.24
Flat roof	0.43	0.02	0.25	0.73
Central AC	0.67	0.88	0.43	0.40
Time to Sale	0.02	0.82	0.47	0.02
Living area	0.88	0.15	0.22	0.92
Parking spaces	0.00	0.26	0.42	0.01
Fireplace	0.57	0.16	0.24	0.37
Waterfront	0.48	0.66	0.31	0.10
Log (median household income)	0.00	0.00	0.54	0.64
Percent white	0.00	0.00	0.38	0.03
Diff at 0.05	9	8	2	4
Diff at 0.10	10	8	3	6

Note: All *P*-values were calculated using the method of seemingly unrelated regressions.

^a Lists the *P*-values for a null of coefficient equality between Pre-Katrina flooded and Post-Katrina non-flooded regressions

^b Lists the *P*-values for a null of coefficient equality between Post-Katrina flooded and Post-Katrina non-flooded regressions

^c Lists the *P*-values for a null hypothesis of coefficient equality between Pre-Katrina flooded and Post-Katrina flooded

^d Lists the *P*-values for a null hypothesis of coefficient equality between Post-Katrina flooded and Post-Katrina non-flooded

in Greater New Orleans prior to Katrina, for flooded and non-flooded areas, respectively. The third and fourth regressions report results post-Katrina, the former relating to the market for flooded houses and the latter for non-flooded. The coefficient estimates represent the percentage change in selling price for a one unit change in each characteristic.

The results across all four equations are consistent with earlier empirical studies. The most common variables have appropriate signs and are significant. Most notable of these common variables are living area, full baths, lot size, age and age squared, central air, fire place, median neighborhood income, and racial composition. Each of these has the expected sign, and is generally significant at the 1% level. Across all the models, the coefficients imply that an increase of 100 square feet of living area increases selling price in a range of from 2.7% to 3.2%, or \$5,293 to \$7,370 at the mean. An additional 1000 square feet in lot size adds from 0.02% to 0.2% to the selling price, or \$40 to \$389 at the mean. For baths, one additional full bath commands a premium ranging from 2.4% to 8.3%, or \$5,528 to \$16,172.

Table 4 Regression results

Dependent variable = Ln(price)	Pre-Katrina Flooded	Pre-Katrina Not flooded	Post-Katrina Flooded	Post-Katrina Not flooded
Elevation(ft)	0.0090*** (0.002)	0.0016 (0.003)	0.0446*** (0.004)	0.0012 (0.002)
Distance to CBD (miles)	-0.0107** (0.004)	-0.0105* (0.006)	-0.0019 (0.008)	-0.0021 (0.005)
Bedrooms	-0.005 (0.006)	0.0340*** (0.012)	0.0183 (0.012)	0.0045 (0.010)
Half-baths	-0.0218*** (0.008)	0.0324** (0.016)	0.0061 (0.014)	0.0459*** (0.013)
Full-baths	0.0505*** (0.009)	0.0481*** (0.016)	0.0825*** (0.016)	0.0243* (0.014)
Age(decades)	-0.782*** (0.007)	0.0045*** (0.008)	0.0068*** (0.012)	0.0062*** (0.008)
Age squared	0.0091*** (0.0011)	0.0045*** (0.0007)	0.0068*** (0.0018)	0.0062*** (0.0009)
Condition	0.1316*** (0.003)	0.1266*** (0.007)	0.1475*** (0.005)	0.1115*** (0.006)
Lot area (1,000 ft ²)	0.0022*** (0.0001)	0.0002*** (0.000)	0.0011*** (0.0003)	0.0009*** (0.0001)
Pool	0.0378*** (0.011)	0.0390* (0.022)	0.0676*** (0.021)	0.0057 (0.018)
Flat roof	-0.1263** (0.049)	0.0532 (0.112)	-0.2623 (0.085)	-0.0225 (0.162)
Central AC	0.0845*** (0.012)	0.0971*** (0.024)	0.0560** (0.024)	0.0628** (0.026)
Time on Mkt(months)	-0.0055*** (0.001)	0.0008 (0.003)	-0.0078*** (0.003)	-0.0070*** (0.002)
Living area (100 ft ²)	0.0308*** (0.001)	0.0313*** (0.002)	0.0266*** (0.002)	0.0317*** (0.002)
Parking spaces	0.0170*** (0.004)	0.0428*** (0.007)	0.0118* (0.006)	0.0210*** (0.006)
Fireplace	0.0416*** (0.008)	0.0509*** (0.0016)	0.0567*** (0.0013)	0.0332*** (0.0013)
Waterfront	0.1430*** (0.053)	0.0791 (0.105)	0.3191*** (0.098)	0.2413*** (0.076)
Ln(MHI)	0.0629*** (0.017)	0.1957*** (0.032)	0.0841*** (0.029)	0.2172*** (0.027)
Percent white	0.0049*** (0.0003)	0.0025*** (0.0005)	0.0043*** (0.0006)	0.0011*** (0.0004)
Constant	10.1098*** (0.215)	9.7035*** (0.298)	10.1863*** (0.310)	9.4054*** (0.194)
Observations	4,464	1,761	1,823	953
R-squared	0.86	0.85	0.81	0.92

Note: Standard errors are in parentheses. Dummy variables for time trend and zip codes were included in regressions, but not reported.

* $P=0.10$

** $P=0.05$

*** $P=0.01$

Holding all else constant, a one percentage point increase in white residents within the census tract raises selling price by 0.1% to 0.5%, or \$253 to \$883 at the mean. Meanwhile, an increase in MHI of 1% results in an increase in price ranging from 0.06% to 0.22%, or \$111 to \$507, when evaluated at the mean. Also, the

presence of a fire place is expected to demand a premium. The estimates for a fireplace range from 3.3% to 5.7%, or \$7,600 to \$11,173.

The age and age squared variables are used to capture the impact of time on the value of housing. It is expected that as a home ages, it will experience a discount relative to other similar homes. At some point this process reverses as old homes become “historic”, and prices reflect a premium. Therefore, the expected sign of age is negative, while the expected sign of age squared is positive. In all four models, this is shown to hold true.

The influence of the other variables changes between models. This suggests that their impacts on selling price differ between the subsets (flood or no-flood) or were altered by the storm (pre-and post-Katrina). In the first two regressions, the distance from the Central Business District (CBD), central air conditioning, and number of parking spaces were all significant, with expected signs. Estimates suggest that each additional mile from the CBD results in a discount of 1.1%, or from \$1,890 to \$2,113 at the mean in the pre-Katrina market, while in both post-Katrina models its effect is not significant. Pre-Katrina, each additional parking space commands a premium ranging from 1.7% to 4.3%. In both post-Katrina regressions, the estimated coefficient on parking is significantly smaller, ranging from 1.2% to 2.1%. Likewise, central air conditioning in the pre-Katrina market commands a premium of from 8.5% to 9.7%, or \$15,016 to \$19,516 at the mean. While still significant in the post-Katrina market, its coefficients are significantly smaller, with estimates ranging from 5.6% to 6.3%. All of these results likely reflect a change in consumers’ priorities. After the storm, residents give priority to living space, or shelter in general, while certain amenities and characteristics do not command the premium they once did.

The assessed condition of the property, not found in most hedonic price models, is critical to this analysis. In this study, it is realized that within the housing market in post-Katrina New Orleans there is a very distinct difference between flooded and non-flooded houses, and between those that are ready for occupants and those consisting of only a shell. These differences are reflected in the scores assigned to each property. The influence of this assessed condition is positive and highly significant in all four models. It is also very consistent across all models.

Prior to Katrina, a one point increase in the agent-assessed condition (on a 5 point scale) results in a premium of 12.7% in areas that did not flood, and 13.2% in areas that flooded. This translates into a premium of \$23,319 to \$25,471, at the mean. After Katrina, the same increase resulted in an 11.2% premium (\$25,680) if the property was not flooded and a 14.8% premium (\$29,011) if the property was flooded. The size of the coefficient, and its stability across regressions, shows that the condition of the structure remains of utmost importance, regardless of the location of the property.

The most revealing results of this analysis are the impacts of those variables related to the risk and extent of flooding, including the elevation of the property’s centroid. Homes at higher elevations are expected to be less likely to flood and/or less likely to suffer significant flooding. Even within a designated flood zone, homes that are located on higher land are considered to be of less risk. Therefore, it is expected that homes at higher elevations will command a premium relative to homes at lower elevations, regardless of flood zone classification.

Prior to Katrina, an additional foot of elevation in flood-prone areas commanded a premium of only 0.9% per foot of elevation, or \$1,590 at the mean. Put differently,

each additional foot below sea-level resulted in a discount of 0.9%. This is consistent with other empirical studies that find a discount associated with locating in a flood zone. However, these earlier studies tend to estimate a singular discount, and ignore the elevation of the property. The discounts found by earlier studies are shown to be roughly equal to the capitalized insurance premiums. This study's estimates seem small, and may illustrate the seemingly worry-free attitude of the residents of New Orleans prior to Katrina, and their confidence in the levy system that protected them. However, it is important to realize that these premia are for each foot of elevation. For example, if an additional 4 feet of elevation would raise an otherwise identical house above that of the other, that difference would command a premium of approximately 4%, which is consistent with the findings of many earlier studies.

After the storm, elevation was at the forefront of many homeowners' minds, which is revealed by the changes in the influence of elevation on property values. For those at-risk (flooded) houses, each foot of elevation commands a 4.5% premium (\$8,821 at the mean), which is consistent with the higher estimates of earlier studies. This is a significant increase from the pre-storm premium. Such an increase is not surprising, as the value of elevation became apparent to everyone in the area.

There are several plausible reasons for this response. One reason may be the difficulties faced by many homeowners in obtaining insurance. In the aftermath of Hurricane Katrina, many insurers stopped writing policies on homes viewed as even slightly risky. Another may simply be a desire to avoid the risk of flooding and the associated costs. Furthermore, the cost of rebuilding is significantly higher in flooded areas due to stricter building codes. Dehring (2006) finds that vacant land values decline with implementation of new, more stringent building codes. Not only are new structures subject to these codes, but any existing structure to which "substantial improvements" are made also fall under these new guidelines. Defined as suffering damage totaling 50% of its appraised value, many of the most heavily damaged homes in New Orleans fall under this requirement. This study finds that houses that were flooded in Katrina displayed a higher sensitivity to future flood risk. That is, once a house was shown to be a flood risk, the question becomes one of extent. For homes that were flooded, the extent of flooding and the associated costs of compliant repairs could be inferred from the high water marks on the buildings. This visible cue likely led to a higher demand and prices for those properties that suffered the least damage, and therefore required lower repair/rebuilding costs.

Houses that were not flooded did not display the same type of sensitivity to elevation. That is, in the eyes of many residents, Katrina seems to have been a once in a lifetime event. If a house did not flood during Katrina, it is unlikely to flood in the future. As a result, there is less incentive to pay for additional elevation. That is not to say that residents do not appreciate the added safety, but additional elevation in non-flooded areas has no impact on selling prices. This perception may have existed prior to Katrina as well. Both before and after the storm, the elevation variable is not significant as a determinant of housing prices for homes already adequately elevated. A possible explanation for this lack of a major impact is that for non-flooded homes, the visual cue of flood-related water marks is missing in areas that did not flood. Another is the diminishing marginal benefit of elevation; homes that did not flood in Katrina were and continue to be unlikely to flood in the future.

Conclusion

The findings of this study highlight two important points about the New Orleans housing market and its responses to flood risks. The first is that residents are aware of, and willing to pay a premium for increased elevation, even if the higher elevation property remains in the same flood zone classification. In effect, the use of flood zone classifications with a range of elevations included assumes that the risks across all homes in that zone are the same. This study shows that a more revealing approach would be to incorporate the actual elevation of each property. While the risk of flooding is important, so is the extent of flooding. Since all properties within a flood zone are not situated at the same elevation, it is reasonable to assume that not all will flood to the same extent. Including the elevations helps illustrate this point.

The second major insight of this study is how the influence of those flood-related characteristics changes in response to a major storm event. Prior to Katrina, the contribution of property elevation was small. However, after Katrina the impact of these variables increased significantly. This indicates that residents began to recognize the value of these characteristics. Several studies (see Speyrer and Ragas 1991 and Bin and Polasky 2004) have shown this to be a common occurrence that may wane in time. But given the media exposure of the plight of New Orleans, this may take much longer than usual.

This study gives rise to several related questions. The current paper and most other studies in the hedonic pricing literature make a parametric assumption regarding the functional form of the relationship between price and flood-risk. Some notable exceptions are Speyrer and Ragas (1991), Pace (1993), and Pace (1995). The most common parametric assumption is that elevation affects price in a log–log or semi-log fashion. Given the large sample size in the current dataset, semi-parametric methods may be used to more carefully examine the relationship between elevation and price. A related issue is whether NFIP guidelines and national flood-insurance rates are properly priced. If this is the case, one would expect monotonicity in the price-elevation function. A useful extension to the present study is to supplement the current hedonic equation with controls for NFIP and a non-parametric elevation function.

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